TECHNOLOGY SOLUTIONS UK LTD
part of HID

www.tsl.com

3166 *BLUETOOTH*® RUGGED UHF RFID RAIN READER

EXTRA-TOUGH, HIGH PERFORMANCE, LONG RANGE UHF RFID READING WITH EPOP-LOQ® CONNECTIVITY AND CHARGING

Supports GEN2X™

RAIN RFID

Bluetooth®

Made for iPhone | iPad





Latest Hardware Advances



The new 3166 RAIN RFID UHF Reader boasts TSL®'s latest hardware innovations and performance improvements, in a more efficient reader with an all-day battery life.

Using the latest generation of Impinj silicon, read rates have been boosted up to 1200 tags per second. Multiple tag operations per tag are possible during an inventory scan - read multiple areas of each tag, or write to or even lock the tag. On reader de-duplication prevents overwhelming the connected host with information.

Support has been added for passive sensor transponders (temperature, strain, humidity, acceleration) from HID and others – contact TSL for information regarding how to utilise these sensor transponders.

Designed to read and write to EPC Class 1 Gen 2 and EPC Class 1 Gen 2V2 tags, the 3166 can also be configured with class leading high performance 2D data scanning to bring unparalleled data collection capabilities to any host it is connected to.

Extensive software support is available for a wide range of platforms including code samples, demonstration applications and source code

Single Point Charge Solution

The 3166 Docking Station allows charging of both the 3166 RAIN RFID UHF Reader and a smartphone or handheld terminal attached via an ePop-Loq mount. The unique design can accommodate a wide range of devices from many handheld and smartphone manufacturers. The 3166 Reader boasts a faster ePop-Loq mode than previous TSL readers to accommodate higher tag throughput.

Tough Impact Resistant Housing

The 3166 Reader from TSL provides high performance UHF RFID reading and writing in an incredibly tough and rugged form factor. The reader is highly resistant to water, dust and mechanical trauma. A high capacity battery provides non-stop operation of the reader over the full working day.

Quick Configuration with Command Scan



3166 Readers with the 2D Imager option can use TSL's patent-pending 'Command Scan' feature to scan special barcodes to configure the Reader. This allows lightning fast Reader configuration, along with on-the-fly adjustment of Reader parameters, opening up a world of improved workflow scenarios. Some barcode examples can be viewed in this document.

Batch Mode

Transponder EPC readings can optionally be stored on the embedded 32GB Micro SD card, meaning that the 3166 Reader can be used independently of a host device. Up to 500 million* EPCs can be stored - all date and time stamped by the on-board Real Time Clock.

Market's Most Advanced Security



The 3128 Reader is future regulation/Cyber Security Acts ready. It features Tag Authentication, Secure Data Storage, Encrypted Communication, Data Acquisition Authenticity checks, Anti-Counterfeiting Measures and Personnel Authentication.

Features:

Faster tag read rate
Gathers RFID data at a blistering pace - up to 1200 tags per second

Longer Operating Times
Up to 34% increase in battery life compared to the previous generation 2166 Reader

Multiple RF Profiles
- high sensitivity
- maximum read rate
- dynamic switching between modes

ETSI Upper Band Option

Bulk encoding feature

***Bluetooth*® improvements**
Around a 2x speed boost for Bluetooth Classic in iOS. Bluetooth Smart, also known as Bluetooth Low Energy, provides a modern, secure link to even more devices, no need for MFI app approval, and out-of-band battery status and device information).

Hardware and OS Independence
Use standalone, or with a wide variety of Bluetooth wireless technology enabled host devices including smartphones, tablets, Enterprise Handhelds and PC's. Operates with Android, iOS, Linux, Mac, Windows.

Backward Compatible
TSL focus on providing long term support of their RFID Readers and minimal design disruption. New TSL Readers are fully compatible with existing software. Software only needs to be updated if wanting to make use of newly added features.

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3166 SPECIFICATIONS

Physical and Environmental Characteristics

Dimensions:	178 x 105 x 172 mm (LxWxH).
Weight:	865 g / 30.5 oz (including battery).
User input:	Single stage trigger.
User feedback:	Speaker, vibration motor, LEDs.
Power:	Rechargeable Lithium Ion removeable battery pack (10.8V, 3.35Ah, 36.2Wh).
Minimum operating time[1]:	Light use[2]: 33.5 hrs Moderate use[3]: 21.5 hrs Heavy use[4]: 12 hrs[5]
Input Rating:	15.0Vdc, 4.34A.
Enclosure materials:	Polycarbonate and TPU.

Performance Characteristics

RFID engine:	TSL custom module with embedded Impinj E710.
Communication protocols:	TSL ASCII 3 parameterised command set.
Memory:	Embedded 32GB storage memory - store up to 500 million date and time stamped EPCs
Compatible Host devices (*Bluetooth*):	Any *Bluetooth* Host[6] supporting the Serial Port Profile (SPP), Human Interface Device (HID), BLE with HOG (HID over GATT) or BLE with SOG (Serial over GATT). See Bluetooth Mode Comparison.
Compatible Host devices (USB):	Any USB host with USB CDC support (Windows, Linux, Mac, Android).

Environmental

Operating Temp.:	-20°C to 55°C (-4°F to 131°F).
Charging Temp.:	5°C to 40°C (41°F to 104°F).
Storage Temp.:	Less than 1 month at at -20 to +60°C (-4°F to 140°F). Less than 3 months at -20°C to +45°C (-4°F to 113°F). Less than 1 year at -20°C to +20°C (-4°F to 68°F).
Humidity:	5% to 85% non-condensing.
Drop Spec:	1.8m.
Tumble:	1500 0.5 metre tumbles at room temperature (3,000 cycles).
Environmental Sealing:	IP67.
Electrostatic Discharge (ESD):	± 15kVdc air discharge; ± 8kVdc contact discharge.
MIL-STD 810F:	Meets and exceeds applicable MIL-STD 810F for drop, tumble and sealing.

RFID Performance

Standards supported:	EPC Class 1 Gen 2 v2
Nominal read range[7]:	Up to 9 m (29.5 ft).
Nominal write range[7]:	Up to 4 m (13.1 ft).
Field:	110-degree forward facing (approx.) measured from front of device.
Antenna:	Circularly Polarised.
Frequency Range:	EU: 865-868MHz, 916-919MHz US: 902-928MHz
Maximum Output Power:	Up to 30 dBm (region dependent) + 4.0 dBiC Antenna.

Barcode Scanning

Optional 2D Barcode Engine:	Optional TSL custom 2D Barcode Scan Engine module.
Sensor Resolution:	1280 x 960 pixels, rolling shutter
Field of View:	Horizontal: 44.5°, vertical: 33.5°
Focal Distance:	From front of engine: 15.24 cm (6 in.)
Aiming LED:	Green LED
Illumination:	1 warm white LED
Symbologies Supported:	1D: All major codes 2D: PDF417, MicroPDF417, Composite, RSS, TLC-39, Datamatrix, QR code, Micro QR code, Aztec, MaxiCode Postal Codes: US PostNet, US Planet, UK Postal, Australian Postal, Japan Postal, Dutch Postal (KIX).
Ranges[8]:	

Barcode	Near	Far
5 mil Code 39	6.1 cm	24.1 cm
5 mil Code 128	7.1 cm	22.9 cm
6.67 mil PDF 417	6.1 cm	20.3 cm
10 mil DataMatrix	7.4 cm	21.6 cm
100% UPCA	4.6 cm	49.5 cm
15 mil QR	3.0 cm	29.2 cm
20 mil QR	3.0 cm	35.6 cm

Communication

Bluetooth:	*Bluetooth* v4.2 compliant (v5.1 compatible)
Bluetooth GATT Services:	• Device Information Service • Battery Service • HID over GATT • Serial over GATT (TSL)
Bluetooth Frequency Range:	2.4 - 2.4835 GHz.
Bluetooth Profiles:	SPP Profile, HID Profile, Apple iAP2, Bluetooth Low Energy.
Bluetooth Range[9]:	Up to 100m.
Bluetooth Pairing:	Simple Secure Pairing, NFC OOB Pairing.
Physical host device connection	Connection via ePop-Loq cases (separate purchase).

[1] Minimum operating time figures are based on new units that have been stored, charged and operated within the stated Environmental Specifications. Units stored over 3 months must be recharged every 3 months. Number of transponders in the environment affects minimum operating time.

[2] Light Use: Continuous RFID inventories for 20s of every 120s

[3] Moderate Use: Continuous RFID inventories for 10s of every 30s

[4] Heavy Use: Continuous RFID inventories for 59s of every 60s

[5] When operating in the "Eco" battery saver mode.

[6] Compatible *Bluetooth* stack required in the Host device

[7] Tag Read/Write performance is dependent on tag type, items tagged, number of tags in the field and other radio and environmental factors

[8] Artificial lighting can affect scanning performance

[9] Open field

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Peripherals and Accessories

External interface:	8-way sealed connector with gold plated contacts.
Bundled items:	Battery.
Other accessories available:	Docking Station with power supply and Mini USB cable. Adapter mounts for a variety of smartphones and handheld terminals.

Regulatory

Regions	EU (CE), USA (FCC), Canada (see page 4 for details)
FCC ID	S6J3166
IC	8948A-3166
EMC	EN 55032: 2015+A11:2020, Class A EN 6100-3-2: 2014, Class A EN IEC 61000-3-2: 2019+A1:2021, Class A EN 61000-3-3: 2013+A1:2019+A2:2021 EN 55035: 2017+A11:2020 EN 301 489-1 V2.2.3 (2019-11) EN 301 489-3 V2.3.2 (2023-01) EN 301 489-17 V3.2.4 (2020-09) 47 CFR FCC Part 15, Subpart B, Class A ANSI C63.4-2014 ANSI C63.4a-2017 ICES-003: 2020 Issue 7, Class A ICES-Gen: 2018 Issue 1 +A1:2021 ANSI C63.4-2014 amended as per ANSI C63.4a-2017
RF	EN 302 208 V3.3.1 (2020-08) 47 CFR FCC Part 15, Subpart C (Section 15.247) ANSI C63.10-2013 Canada RSS-247 Issue 3, August 2023 Canada RSS-Gen Issue 5, Amendment 2, February 2021 ANSI C63.10-2013
RF Exposure	EN 50566:2017 EN 62209-2:2010/A1:2019 IEC 62209-2:2010/AMD1:2019 EN 50663:2017 EN 62479:2010 EN 50364:2018 EN 62369-1:2009 IEEE Std 1528:2013 KDB 865664 D01 v01r04 KDB 865664 D02 v01r02 KDB 447498 D01 v06 ISED RSS-102 Issue 5:2015/AMD1:2021 IEC/IEEE 62209-1528:2020
Electrical Safety	IEC 62368-1:2018 UL 62368-1:2019 CSA C22.2 No. 62368-1:19
Environmental	IEC 60529 Edition 2.2 2013-08 (IP67) **2011/65/EU (RoHS 2)** Restriction of the use of certain Hazardous Substances in electrical and electronic equipment **2015/863 (RoHS 3)** Amendment to Annex II of 2011/65/EU

Warranty

The TSL 3166 reader is warranted against manufacturing defects for a period of one year (12 months), provided the product remains unmodified and is operated under normal and proper conditions.

Full warranty information can be downloaded from the TSL website at www.tsl.com/warranty.

Terms

"Made for iPod," "Made for iPhone," and "Made for iPad" mean that an electronic accessory has been designed to connect specifically to iPod, iPhone, or iPad, respectively, and has been certified by the developer to meet Apple performance standards. Apple is not responsible for the operation of this device or its compliance with safety and regulatory standards. Please note that the use of this accessory with iPod, iPhone, or iPad may affect wireless performance.

iPad, iPhone, iPod and iPod touch are trademarks of Apple Inc., registered in the U.S. and other countries.

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TSL RFID Apps

These TSL RFID Demo apps have all been written using our free Software Development Kits:


RFID Sensor Tag Explorer
https://www.tsl.com/apps/sensor-tag-explorer


RFID Scan Scan Write
www.tsl.com/apps/rfid-scan-scan-write


RFID Tag Finder
www.tsl.com/apps/rfid-tag-finder


RFID Web Wedge
www.tsl.com/apps/rfid-web-wedge


RFID Explorer
www.tsl.com/apps/rfid-explorer


TSL Reader Configuration
www.tsl.com/apps/tsl-reader-configuration


RFID Inventory
https://www.tsl.com/apps/rfid-inventory

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3166 PART NUMBERS

Countries	Part Numbers	Notes
ETSI Lower Band Albania, Andorra, Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Falkland Islands, Finland, France, French Guiana, Georgia (Licence Required), Germany, Greece, Greenland, Guernsey, Guadeloupe, Hungary, Iceland, Ireland, Italy, Jersey, Latvia, Liechtenstein, Lithuania, Luxembourg, Macedonia, Malta, Martinique, Monaco, Montenegro, Netherlands, Norway, Poland, Portugal, Romania, Saudi Arabia, Slovakia, Slovenia, Spain, Sweden, Switzerland, UAE, United Kingdom	With 2D barcode imager: 3166-ES1-EU No barcode imager: 3166-EX1-EU	• ETSI Lower Band • Operating frequency: 865 – 868 MHz
ETSI Upper Band Albania, Austria, Belgium*, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Greece, Hungary, Iceland, Ireland, Italy, Liechtenstein*, Lithuania, Luxembourg, Macedonia, Malta, Moldova, Montenegro, Norway, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland*, Türkiye, United Kingdom *Limited implementation	With 2D barcode imager: 3166-AS1-EU No barcode imager: 3166-AX1-EU	• ETSI Upper Band • Operating frequency: 916 - 919MHz • Each country listed **may have further requirements or restrictions** - for more information, visit www.tsl.com/uhf-frequency-allocations/
FCC Bangladesh, Canada, Ecuador, El Salvador, Guam, Guatemala, Northern Mariana Islands, Puerto Rico, United States of America	With 2D barcode imager: 3166-AS1-US No barcode imager: 3166-AX1-US	• FCC • Operating frequency: 902 - 928 MHz
Australia and New Zealand	3166-AS1-AU-NZ 3166-AX1-AU-NZ	• Operating frequency: 920 - 925 MHz
Brazil	3166-AS1-BR 3166-AX1-BR	• Operating frequency: 902 - 907, 915 - 927 MHz
Colombia	3166-AS1-CO 3166-AX1-CO	• Operating frequency: 915 - 927 MHz
Mexico	3166-AS1-MX 3166-AX1-MX	• Operating frequency: 902 - 928 MHz
Qatar	3166-ES1-QA 3166-EX1-QA	• Operating frequency: 865 – 868 MHz

If you are interested in purchasing for a country/region that is not listed above, please contact enquiries@tsl.com for assistance.

Accessories	Part Numbers
1166/3166 Docking Station Kit, 65W PSU and Mini USB lead	1166-CRD-01-KIT
1166/3166 External Battery Charger. Includes PSU and UK Line Cord	1166-BC-UK
1166/3166 External Battery Charger. Includes PSU and US Line Cord	1166-BC-US
1166/3166 External Battery Charger. Includes PSU and EU Line Cord	1166-BC-EU

Accessories	Part Numbers
1166/3166 External Battery Charger. Includes PSU and AUS/NZ Line Cord	1166-BC-AU-NZ
Spare Battery - Rechargeable Lithium Polymer for 1166/3166 UHF Reader	1166-00-BA-3000
Line Cord (UK Plug, 2m)	IEC-2M-UK
Line Cord (US Plug, 1.8m)	IEC-1.8M-US
Line Cord (EU Plug, 1.8m)	IEC-1.8M-EU
Line Cord (AU/NZ Plug, 2m)	IEC-2M-AU-NZ

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MOUNTING A HOST DEVICE ON THE UHF READER

You can physically connect Enterprise Hand-Held Terminals to your UHF RFID Reader using ePop-Loq® mounts. These custom made mounts can provide direct USB charge and data connections (when available) between the Reader and host device. To enquire about ePop-Loq mounts for devices not listed below, please contact enquiries@tsl.com.


Datalogic Memor 10/11




Datalogic Memor 12/17




Datalogic Memor 30/35




Honeywell CT40/45/45XP/47




Honeywell CT50/60




Honeywell D75e




Honeywell EDA50/51




Janam XT30/40




Zebra TC21/26




Zebra TC22/27




Zebra TC51/52/56/57




Zebra TC70/72/75/77



Device Mount	Part Numbers
ePop-Loq Mount for Datalogic Memor 10/11	2108-A-EPL-CASE
ePop-Loq Mount for Datalogic Memor 12/17	2118-A-EPL-CASE
ePop-Loq Mount for Datalogic Memor 30/35	2117-A-EPL-CASE
ePop-Loq Mount for Honeywell CT40, CT45/ CT45XP and CT47	2103-A-EPL-CASE-V3
ePop-Loq Mount for Honeywell CT50/CT60	1192-A-EPL-CASE
ePop-Loq Mount for Honeywell D75e	1190-A-EPL-CASE
ePop-Loq Mount for Honeywell EDA50	1195-C-EPL-CASE
ePop-Loq Mount for Honeywell EDA51	1217-C-EPL-CASE
ePop-Loq Mount for Janam XT30/XT40	ATC-XT-01 (order through Janam Technologies)
ePop-Loq Mount for Xiaomi Mi A2	2106-A-EPL-CASE
ePop-Loq Mount for Zebra TC20/TC25	2101-A-EPL-CASE
ePop-Loq Mount for Zebra TC21/TC26	2120-A-EPL-CASE
ePop-Loq Mount for Zebra TC22/TC27	2119-A-EPL-CASE
ePop-Loq Mount for Zebra TC51/52/56/57	1199-A-EPL-CASE
ePop-Loq Mount for Zebra TC70/72/75/77	1198-A-EPL-CASE



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ABOUT TSL

Technology Solutions UK Ltd (TSL®), part of HID, is a leading manufacturer of high performance mobile RFID readers used to identify and track products, assets, data or personnel.

For over two decades, TSL has delivered innovative data capture solutions to Fortune 500 companies around the world using a global network of distributors and system integrators. Specialist in-house teams design all aspects of the finished products and software ecosystems, including electronics, firmware, application development tools, RF design and injection mould tooling.

TSL is an ISO 9001:2015 certified company.

ISO 9001: 2015

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ABOUT HID

HID powers the trusted identities of the world's people, places and things. We make it possible for people to transact safely, work productively and travel freely. Our trusted identity solutions give **people** convenient access to physical and digital **places** and connect **things** that can be identified, verified and tracked digitally. Millions of people around the world use HID products and services to navigate their everyday lives, and billions of things are connected through HID technology. We work with governments, educational institutions, hospitals, financial institutions, industrial businesses and some of the most innovative companies on the planet. Headquartered in Austin, Texas, HID has over 4,000 employees worldwide and operates international offices that support more than 100 countries. HID is an ASSA ABLOY Group brand.

For more information, visit www.hidglobal.com.

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